

Loops: The Lucky Dice Simulator

Overview

In this assignment, you will write a program that simulates rolling a dice until you get a 6. This will practice all the flow control concepts you learned in Chapter 3. To get started, ensure that you download the template file attached to this assignment in Blackboard. Use the "Load" button in Mu to load the attached template file and start your work. Do note that double-clicking a python file will attempt to run the file. This can make it appear as though the file opened and closed quickly but did nothing. In fact, the program ran and closed in fractions of a second. If you want to edit a file, you must load it into an editor as previously described. You should only make the necessary changes to the template based on the TODO comments.

Objectives

- To create a functional dice simulator game.
- To practice writing and using loops in Python.
- To handle user input and implement basic if-else logic.
- To reinforce concepts of user interaction and flow control in Python programming.

Requirements

- Your program should keep asking the user to roll a dice until they roll a 6.

Steps to Implement

1. Import the 'random' module.
2. Create an infinite loop using 'while True'.
3. Prompt the user for input as follows: "Type 'roll' to play or 'done' to exit: "
4. If they type 'done', break the loop.
5. If they type anything other than 'roll' or 'done', print "Invalid command. Please try again." and restart the loop using the 'continue' statement.
6. If they type 'roll', print "Rolling..." 3 times using a 'for' loop.
7. Generate a random number between 1 and 6
8. If they roll a 6, print "You rolled a six! You win!" and break the loop.
9. Print "Game Over"